

Magic matrices for ionization in mass spectrometry



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ABSTRACT

The initial discovery that a heated inlet tube of a mass spectrometer is an ionization source producing ions from volatile, nonvolatile, small, and large molecules with charge states similar to electrospray ionization has been advanced to ionization requiring only the vacuum inherent with a mass spectrometer and a suitable matrix. This spontaneous ionization method was first applicable with the matrix 3-nitrobenzonitrile. Here we report that over 40 compounds have now been discovered that spontaneously convert molecules to gas-phase ions when exposed to sub-atmospheric pressure, some with remarkable sensitivity (10 fmol of protein insulin). The commonality of all matrices is the ability to sublime, preferably near room temperature, through exposure to vacuum, and the ability to create charge separation under these conditions. The effect of vacuum, airflow, temperature (−80 to +150 °C) and pH (1–9) on the effectiveness of these newly discovered matrices to ionize peptides and proteins is presented. Compounds with and without acidic hydrogen atoms act as matrices and ionize specific compound classes. The new matrices extend applications from peptides, proteins and drugs to compound classes without basic functionality such as lipids and synthetic polymers in the negative and positive modes. Mass resolution and ion mobility spectrometry aspects are also discussed.

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1. Introduction

During the 50 years covered by this special issue, great strides were made in learning how to extend mass spectrometry (MS) to high-mass compounds. In these efforts, the use of solvents or small-molecule matrices were the most important factors for success. Fast atom bombardment by Barber and coworkers [1] introduced liquid matrices, and matrix-assisted laser desorption/ionization (MALDI) by Karas and Hillenkamp [2] introduced small molecule solid matrices for use in MS. In any method capable of ionizing large molecules such as proteins, either high voltage as in electrospray ionization (ESI) [3] or high energy input [4,5] were employed. Recently, we introduced a method in which only the energy provided by the vacuum is necessary to spontaneously convert small or large molecules into gas-phase ions upon exposure of a matrix containing analyte to the vacuum of the mass spectrometer [6–9].

This discovery originated from a systematic study that led to the discovery of laserspray ionization (LSI), a method in which laser ablation of a common MALDI matrix, 2,5-dihydroxybenzoic acid (2,5-DHB), at atmospheric pressure produced multiply charged ions from peptides and proteins [10,11]. It was initially assumed that the laser was essential for ionization, however, the discovery that the laser is only a means of ablating matrix–analyte into the heated inlet tube of the mass spectrometer where ionization is initiated led to the term matrix assisted ionization *inlet* (MAII) [12,13]. In MAII, matrix–analyte sample on a substrate can be dislodged by tapping against the inlet aperture in a fashion that particles enter the heated inlet tube linking atmospheric pressure and the vacuum of the mass analyzer producing ions from incorporated analyte. A solution–analyte mixture can substitute for the solid matrix–analyte and similarly produce analyte ions in the inlet [14,15]. In this solvent assisted ionization *inlet* (SAII) method, as with LSII and MAII, the heated inlet is the ion source. These *inlet* ionization methods produce charge states nearly identical to ESI. Because the inlet tube was heated in these experiments [16], it was postulated that charged particles/droplets are produced by a thermal process and upon desolvation produce the observed bare ions [17,18]. Types of gases, higher initial

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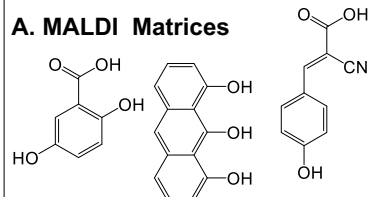
pressure, downstream pressure, rf voltages, and collision with surfaces have been parameters shown to enhance analyte ion formation [13,16,18–20].

The finding that similar charge states are produced by laser ablation in a vacuum MALDI source using select matrix

compounds in the absence of conductive heat led to the terminology of laserspray ionization *vacuum* (LSIV) [19]. In LSIV a heated inlet tube is not required. It was postulated that laser ablation of the matrix provided the thermal energy necessary to produce the charged droplets. Analyte ions are observed so long as

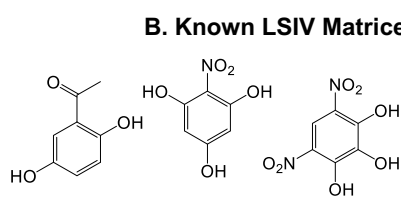
I. Matrix Structures for *Vacuum* Ionization using an Intermediate Pressure MALDI Source

A. MALDI Matrices



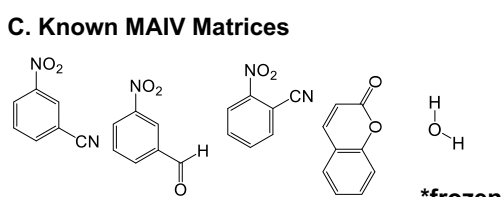
2,5-DHB dithranol CHCA

B. Known LSIV Matrices



2,5-DHAP 2-NPG DNPG

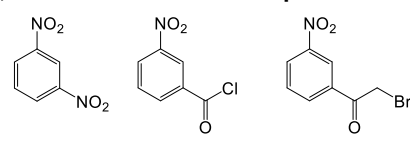
C. Known MAIV Matrices



*3-NBN 3-NBZ 2-NBN *coumarin *frozen H₂O

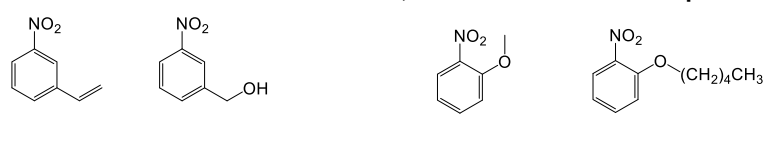
II. Novel MAIV Matrices

A. 1,3-Substituted Nitro Compounds



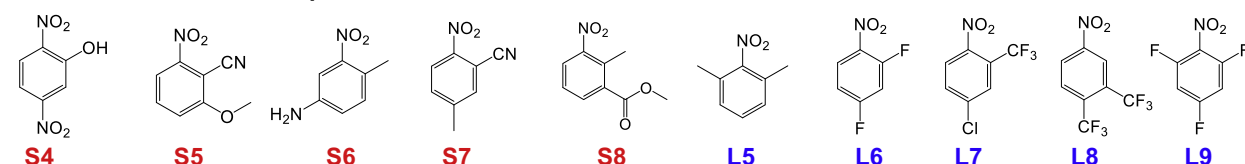
S1 S2 S3

B. 1,2-Substituted Nitro Compounds



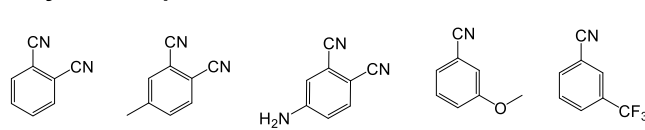
L1 L2 L3 L4

C. Multi-Substituted Nitro Compounds



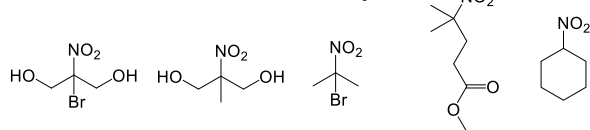
S4 S5 S6 S7 S8 L5 L6 L7 L8 L9

D. Cyano Compounds



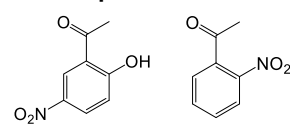
S9 S10 S11 L10 L11

E. Linear and Non-Aromatic Compounds



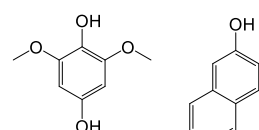
S12 S13 L12 L13 L14

F. Acetophenones



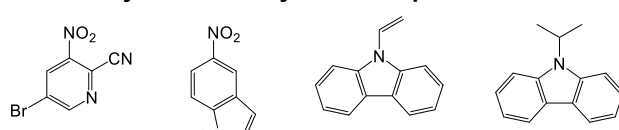
S14 L15

G. Alcohols



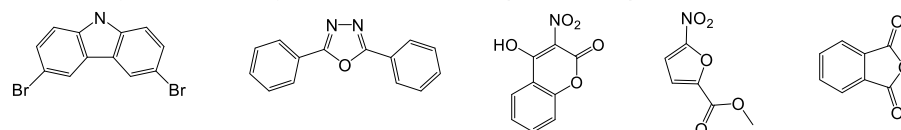
S15 *S16

H. Heterocyclic and Anhydride Compounds



S17 S18 *S19 *S20

H. Heterocyclic and Anhydride Compounds (continued)



*S21 *S22 *S23 S24 *S25

S: solid at room temperature
L: liquid at room temperature
***:** Known triboluminescence compounds

Scheme 1. (I) Structures of known matrix compounds for (A) MALDI, (B) LSIV, and (C) MAIV. (II) Structures of novel MAIV matrix compounds arranged by: (A) 1,3-substituted nitro compounds: 1,2-dinitrobenzene (S1), 3-nitrobenzoyl chloride (S2), 2-bromo-3'-nitroacetophenone (S3), 3-nitrostyrene (L1), 3-nitrobenzyl alcohol (L2); (B) 1,2-substituted nitro compounds: 2-nitroanisole (L3), 2-nitrophenyl pentyl ether (L4); (C) multi-substituted nitro compounds: 2,5-dinitrophenol (S4), 6-nitro-*o*-anisnitrile (S5), 3-methyl-4-nitroaniline (S6), 5-methyl-2-nitrobenzonitrile (S7), methyl-2-methyl-3-nitrobenzoate (S8), 1,3-dimethyl-2-nitrobenzene (L5), 2,4-difluoronitrobenzene (L6), 4-chloro-1-nitro-2-(trifluoromethyl) benzene (L7), 3,4-bis(trifluoromethyl) nitrobenzene (L8), 3,4,5-trifluoronitrobenzene (L9); (D) cyano compounds: phthalonitrile (S9), 4-methyl-phthalonitrile (S10), 4-aminophthalonitrile (S11), 3-methoxybenzonitrile (L10), 3-(trifluoromethyl)-benzonitrile (L11); (E) linear and non-aromatic compounds: 2-bromo-2-nitro-1,3-propanediol (S12), 2-methyl-2-nitro-1,3-propanediol (S13), 2-bromo-2-nitropropane (L12), methyl-4-methyl-4-nitropentanoate (L13), nitrocyclohexane (L14); (F) acetophenones: 2'-hydroxy-5'-nitroacetophenone (S14), 2'-nitroacetophenone (L15); (G) alcohols: 1,4-dihydroxy-2,6-dimethoxybenzene (S15), 2-naphthol (*S16); (H) heterocyclic and anhydride compounds: 5-bromo-3-nitropyridine-2-carbonitrile (S17), 5-nitroindole (S18), 9-vinyl carbazole (*S19), 9-isopropyl-9H-carbazole (*S20), 3,6-dibromocarbazole (*S21), 2,5-diphenyl-1,3,4-oxadiazole (*S22), 4-hydroxy-3-nitrocoumarin (*S23), methyl 5-nitro-2-furoate (S24), phthalic anhydride (*S25). Matrices are numbered and designated by physical state at room temperature (solid, red = S and liquid, blue = L). *Indicates the matrix compound is reported to triboluminesce (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

collisions with gas-phase molecules or surfaces provide desolvation of the matrix. Without a source of energy other than the laser, *vacuum* matrices are required to be more volatile than those used in *inlet* ionization, where heat can be supplied over a longer time.

A major impetus of the work reported here is to discover additional matrix compounds in order to better understand properties that are important for a matrix that produce ESI-like ions from the solid (MAII) or liquid (SAII) state without the conventional high energy input, and how these matrices can improve the analytical utility of the method. Initial studies focused on finding matrices to produce multiply charged ions using MAII [13]. Without the need of a laser, but with the inlet tube of a Thermo Scientific LTQ Velos mass spectrometer heated to 450 °C, 137 of 176 small molecule compounds tested worked as MAII matrices, ionizing analytes such as ubiquitin when the matrix–analyte placed on a glass slide was tapped against the inlet tube and physically introduced into the inlet apertures in crystalline form. Most of the compounds that failed as MAII matrices are liquids at room temperature. Interestingly, several of the solid matrices that also worked poorly or failed at an inlet tube temperature of 450 °C are traditional MALDI matrices such as sinapinic and α -cyano-4-hydroxycinnamic acid (CHCA) (Scheme 1IA). Out of the compounds that work as *inlet* ionization matrices, no common chemical structural element was found to be important. While a large number of compounds serve as matrices with a hot inlet tube, most mass spectrometers are not so equipped. Amazingly, 5 of these matrices produced ions from ubiquitin and insulin with an inlet tube temperature of only 50 °C.

A search for more volatile matrices led to the discovery of the matrix compound 3-nitrobenzonitrile (3-NBN) which produces analyte ions upon exposure to vacuum conditions by default available with any mass spectrometer [6–9]. 3-NBN has unusual properties of both triboluminescence [21] and sublimation [22]. The internal energy of this matrix can be elevated at atmospheric pressure by exposure to moderate heat, thereby increasing the rate of analyte ion production [9,23]. In analogy to MAII, this method in which a matrix–analyte spontaneously produces ions when presented to sub-atmospheric pressure conditions is termed matrix assisted ionization *vacuum* (MAIV). In MAIV, the vacuum at the inlet aperture of an atmospheric pressure mass spectrometer is sufficient to initiate ionization [6,7,9]. Without the need of a heated inlet for ionization, skimmer cone based inlet apertures such as found with the Waters Z-Spray design for ESI, as well as intermediate pressure MALDI sources are applicable with MAIV [6–9]. Even mild heat outside the inlet tube allows to initiate MAIV [23]. Sensitivities of 5 fmol have been achieved for small molecule clozapine and 50 fmol for protein ubiquitin in the intermediate pressure vacuum source of a Waters SYNAPT G2 [6,7].

Here we show that MAIV is a general ionization process and that different matrices ionize specific compound classes, similar to MALDI [24–48]. New approaches for screening compounds relative to their utility as matrices are provided, as is our assessment of the criteria for a successful matrix. Analyte ion abundance relative to temperature and solution pH is studied for 25 new and 4 previously discovered MAIV matrices. With the proper matrix and vacuum assistance, analyte ions are produced at temperatures as low as ~200 K.

2. Experimental

2.1. Materials

Matrices were selected on the expectation or knowledge that they sublime in vacuum near room temperature [22,49–55] and some were selected because they are known to exhibit triboluminescence

[56–68]. The lettering “S” indicates the compound is a solid at room temperature and “L” indicates the compound is a liquid at room temperature (Scheme 1). Matrices and analytes were used as purchased for screening purposes; a few matrices were recrystallized. Matrix compounds 3-methyl-4-nitroaniline (S6) and 2-methyl-2-nitro-1,3-propanediol (S12) as well as polyethylene glycol (PEG) 1000 standard were purchased from Alfa Aesar (Ward Hill, MA, USA). Matrix compounds 3-nitrobenzonitrile (3-NBN), coumarin, 1,3-dinitrobenzene (S1), 3-nitrobenzoyl chloride (S2), 2-bromo-3'-nitroacetophenone (S3), 2,5-dinitrophenol (S4), 6-nitro-*o*-anisonitrile (S5), 5-methyl-2-nitrobenzonitrile (S7), methyl-2-methyl-3-nitrobenzoate (S8), phthalonitrile (S9), 4-methyl-phthalonitrile (S10), 4-aminophthalonitrile (S11), 2'-hydroxy-5'-nitroacetophenone (S14), 1,4-dihydroxy-2,6-dimethoxybenzene (S15), 2-naphthol (S16), 5-bromo-3-nitropyridine-2-carbonitrile (S17), 5-nitroindole (S18), 9-vinyl-carbazole (S19), 9-isopropyl-9H-carbazole (S20), 3,6-dibromocarbazole (S21), 4-hydroxy-3-nitrocoumarin (S23), methyl-5-nitro-2-furoate (S24), phthalic anhydride (S25), 3-nitrostyrene (L1), 3-nitrobenzylalcohol (L2), 2-nitroanisole (L3), 2-nitrophenyl pentyl ether (L4), 1,3-dimethyl-2-nitrobenzene (L5), 2,4-difluoronitrobenzene (L6), 4-chloro-1-nitro-2-(trifluoromethyl) benzene (L7), 3,4-bis(trifluoromethyl) nitrobenzene (L8), 3,4,5-trifluoronitrobenzene (L9), 3-methoxybenzonitrile (L10), 3-(trifluoromethyl)-benzonitrile (L11), 2-bromo-2-nitropropane (L12), methyl-4-methyl-4-nitropentanoate (L13), nitrocyclohexane (L14), and 2'-nitroacetophenone (L15) as well as insulin (bovine pancreas), ubiquitin (bovine), disialoganglioside (GD1_a) from bovine brain, clozapine, and sodium trifluoroacetate salt (NaTFA) were purchased from Sigma–Aldrich. 3-NBN was also obtained from TCI America (Portland, OR, USA). 2-Bromo-2-nitro-1,3-propanediol (or bronopol, S12), 2,5-diphenyl-1,3,4-oxadiazole (S22), and 2,5-dihydroxyacetophenone (2,5-DHAP) were purchased from Acros Organics (Pittsburgh, PA, USA). The peptides, leucine enkephalin, angiotensin I and angiotensin II were obtained from American Peptide (Sunnyvale, CA, USA). Acetonitrile, methanol, petroleum ether, formic acid (FA), HPLC grade water, and 10 μ L pipet tips were obtained from Fisher Scientific (Pittsburgh, PA, USA). Glacial acetic acid and ammonium hydroxide (NH₄OH) were purchased from Mallinckrodt Chemical (Phillipsburg, NJ, USA). Hydrochloric acid (HCl) was obtained from EMD (Billerica, MA, USA) and ethanol from Decon Labs (King of Prussia, PA, USA). Glass microscopy slides were purchased from Gold Seal (Portsmouth, NH, USA). Microscopy images of matrix crystals were taken with a Nikon Eclipse LV100 optical microscope. A biodryer was purchased from BioSpec (Bartlesville, OK, USA) and TissueLyser from Qiagen (Hilden, Germany).

2.2. Recrystallization

All compounds that produced ions by MAIV were recrystallized except for 3-nitrobenzoyl chloride (S2), 6-nitro-*o*-anisonitrile (S5), 5-methyl-2-nitro-benzonitrile (S7), and 9-isopropyl-9H-carbazole (S20). Matrices were recrystallized from methanol or ethyl acetate and washed with ice cold water or methanol over suction filtration. 2,5-Diphenyl-1,3,4-oxadiazole (S22) was recrystallized twice in ethanol and petroleum ether and washed with ice cold water to obtain the differing crystal morphologies of large plates (type II) and fine needles (type I) as described [66].

2.3. Sample preparation and introduction to mass spectrometer inlet systems

Stock solutions of leucine enkephalin, angiotensin I, angiotensin II, and ubiquitin were prepared as 1 mg mL⁻¹ in pure

water and bovine insulin as 1 mg mL^{-1} in 50:50 acetonitrile:water with 0.1% acetic acid. The stock solutions were then mixed and diluted in pure water for a final concentration of $5 \text{ pmol } \mu\text{L}^{-1}$ for all standards. Polymer PEG 1000 was prepared as a $20 \text{ pmol } \mu\text{L}^{-1}$ solution in 50:50 acetonitrile:water and mixed with NaTFA salt in a molar ratio of 1:10 analyte:salt. GD1_b lipid was used for negative mode measurements and prepared as a 1 mg mL^{-1} stock solution in methanol and diluted to $5 \text{ pmol } \mu\text{L}^{-1}$ in 50:50 methanol:water. Angiotensin I and ubiquitin were used in a pH study and diluted to $10 \text{ pmol } \mu\text{L}^{-1}$ in 50:50 methanol:water containing 0.1% HCl, 0.1% FA, 50:50 methanol:water, or 0.1% NH_4OH to obtain a pH of 1, 3, 5, and 9, respectively. Matrix and analyte solutions were mixed in molar ratios of matrix:analyte ranging from 2.1×10^4 :1 to 6.7×10^4 :1 before analysis, resulting in 5 pmol of angiotensin I or ubiquitin consumed per acquisition. The matrix–analyte mixture was introduced to the vacuum of the mass spectrometer in $1 \mu\text{L}$ increments either by drawing the solution into a $10 \mu\text{L}$ pipet tip or spotting onto a glass microscopy slide and allowing the matrix–analyte to crystallize. Dried matrix with incorporated analyte was exposed to the vacuum at the inlet aperture of a Waters SYNAPT G2 mass spectrometer. The source block temperature was used at the optimal temperature for each matrix compound as determined by the positive mode temperature studies. A more detailed description of matrix preparation and differentiation between pipet tip and glass plate sample introduction can be found in the Supplementary data in Table S1.

For solvent-free preparation, $10 \mu\text{L}$ of 1 mg mL^{-1} angiotensin I in water was placed in vials with 3 stainless steel beads. The vials were then placed in a biodryer and solvent was allowed to evaporate over 3 h. Matrix 2,5-diphenyl-1,3,4-oxadiazole (S22) was recrystallized in ethanol and petroleum ether and added to separate vials and the matrix–analyte was mixed using a TissueLyser according to published procedures [69].

Analyses using liquid matrix compounds were prepared using $100 \mu\text{L}$ of a $5 \text{ pmol } \mu\text{L}^{-1}$ solution of angiotensin I in 50:50 acetonitrile:water mixed with $10 \mu\text{L}$ of the liquid matrix solution and vortexed for 60 s. The liquid matrix–analyte solutions were left in a biodryer for 6–12 h until the volume was reduced from $110 \mu\text{L}$ to approximately $10 \mu\text{L}$, indicating all acetonitrile–water solvent was evaporated. The mixtures were then vortexed for 60 s and $2\text{--}3 \mu\text{L}$ of the liquid matrix–analyte solution was drawn into a pipet tip and deposited onto dry ice (-80°C CO_2) before introduction to the SYNAPT G2 Z-Spray source with the source block temperature at 30°C .

2.4. Instrumentation

MAIV was performed on a Waters SYNAPT G2 (Waters Corporation, Milford, MA) from atmospheric and intermediate pressure using commercial Z-Spray and intermediate pressure vacuum MALDI sources, respectively. The atmospheric pressure ESI source housing was removed and interlocks overridden as previously described [7,9]. A commercial Z-Spray source cone was used for matrix–analyte introduction by pipet tip [9]. A modified cone with a widened inlet of ca. 3 mm was used as previously described [7] for matrix–analyte introduction by a glass microscopy slide. The atmospheric pressure source was operated with the source block temperature varied from 30 to 150°C . The intermediate pressure vacuum source ($\sim 0.16 \text{ Torr}$) was operated at ambient temperature with ion extraction near 0 V and the laser turned off, so-called as LSIV settings as previously described [19]. Matrix–analyte samples were spotted onto stainless steel MALDI target plates before inserting into the intermediate pressure source of the SYNAPT G2 mass spectrometer. The mass spectral and drift time data from the SYNAPT G2 were analyzed by MassLynx v4.1 and DriftScope v2 and resolution values were obtained from ResCalc v2.2.3. Microscopy images of

recrystallized matrix compounds were taken with a Nikon Eclipse LV100 optical microscope.

3. Results

3.1. Novel MAIV matrices

The potential importance of a sensitive spontaneous ionization method that only requires a matrix and the vacuum inherent with any mass spectrometer led us to retest all matrix compounds found to produce ions at low inlet tube temperature on the LTQ Velos [13] or intermediate pressure vacuum source of the SYNAPT G2 [7]. Further, the search was widened to test new compounds reported to sublime in vacuum [22,49,51–55] or triboluminesce [56–59, 61–68]. However, with this spontaneous ionization approach, only a single matrix–analyte sample can be tested at a time using a vacuum source [7]. Some compounds entirely sublimed in the two minutes between being exposed to vacuum and reaching the ionization region. Others sublimed slowly without producing any analyte ions. Because of the disadvantages of using the intermediate pressure vacuum source designed for MALDI operation for compound screening, we developed two new approaches to more rapidly test compounds as potential matrix candidates using the atmospheric pressure Z-Spray inlet aperture of the SYNAPT G2 mass spectrometer.

In one method, a device with a larger inlet aperture was used that replaced the original Z-Spray source cone, as previously described [6–9], and is capable of holding a glass microscope slide by the pressure differential between the vacuum of the mass spectrometer and atmospheric pressure. To prevent the mass spectrometer from venting with this arrangement, the isolation valve could be in the open position only when a glass plate sealed the aperture. Samples can be changed in seconds by opening and closing the isolation valve in conjunction with changing the glass slide. Using this device with the ion source at 80°C , 10 compounds (3-nitrobenzoyl chloride (S2), 2,5-dinitrophenol (S4), 5-methyl-2-nitrobenzonitrile (S7), methyl-2-methyl-3-nitrobenzoate (S8), 2-bromo-2-nitro-1,3-propanediol (S12), 2-methyl-2-nitro-1,3-propanediol (S13), 1,4-dihydroxy-2,6-dimethoxybenzene (S15), 4-hydroxy-3-nitrocoumarin (S23), methyl-5-nitro-2-furoate (S24), phthalic anhydride (S25)) were discovered that produce multiply charged ions from a mixture of peptides and proteins (Table S1.I) in addition to the previously discovered MAIV matrices (Table S1.II). Different crystal morphologies are shown for the novel (Table S2.I) as well as known MAIV matrices (Table S2.II). Using microscopy, without applying vacuum or heat, 8 of the successful MAIV matrices (1,3-dinitrobenzene (S1), 2,5-dinitrophenol (S4), methyl-2-methyl-3-nitrobenzoate (S8), 9-vinyl carbazole (S19), 9-isopropyl-9H-carbazole (S20), methyl-5-nitro-2-furoate (S24), 3-NBN and coumarin) display visible sublimation at room temperature over a period of 24 h (Table S2).

A second method, as previously described [9], was found to be efficient for screening purposes and also for rapidly acquiring data on samples brought to our laboratory for analysis. In this approach (Supplementary Movie 1), one microliter of the matrix solution containing the analyte is drawn into a pipet tip and briefly allowed to crystallize followed by the tip being exposed to the vacuum at the unmodified skimmer cone of the commercial Z-Spray source of the SYNAPT G2. Airflow in the region where the pipet tip is brought near the entrance of the Z-Spray source draws matrix–analyte crystals into the vacuum. More compounds are applicable as matrices with this approach, which in general is simpler and produces greater ion abundances than with the glass plate method (Table S1). Airflow and airflow turbulences may aid the ionization process, possibly by assisting sublimation of the matrix either in the initial charge separation process lifting the matrix–analyte from the respective

substrate or with desolvation of the subsequently produced charged particles. This would suggest that higher vacuum might impede sublimation and the ionization process.

For most matrices discovered using the pipet tip approach, a “good” sample is often when the matrix–analyte starts to form the initial crystals, commonly just outside of the pipet tip. Matrix–analyte samples that are left to dry in open air, sometimes even

after a few minutes, tend to produce less analyte ions than those that are slightly wet, especially using the glass plate method. Some matrices that are too wet have not yet sufficiently formed the necessary crystalline sample and often do not produce any analyte ions. A notable exception is methyl 5-nitro-2-furoate (S24) (Scheme 1), which produces highly charged ions of the analyte bovine insulin (MW 5730) when introduced to the inlet as a dried

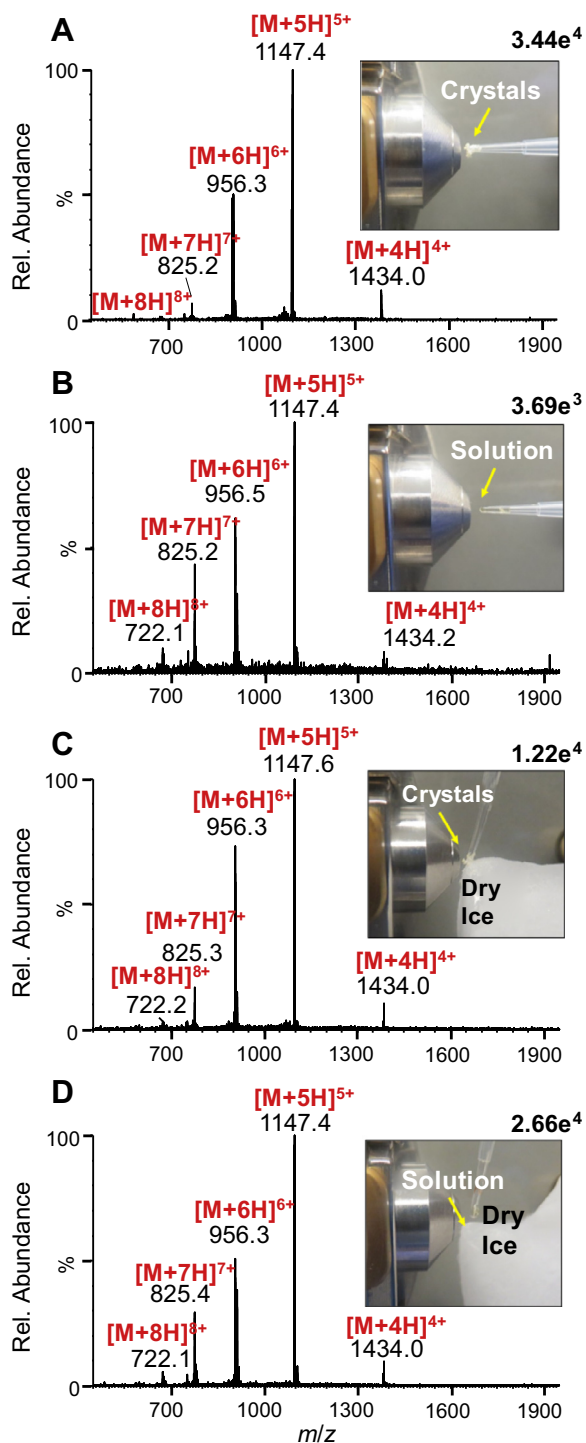


Fig. 1. Methyl-5-nitro-2-furoate (S24) as matrix containing bovine insulin (5 pmol μL^{-1}) introduced to the atmospheric pressure Z-Spray ion source by pipet tip when (A) completely dry (crystalline), (B) in solution, (C) frozen after completely drying (crystalline), and (D) by frozen solution. Insets show pictures of introduction of matrix–analyte into the atmospheric pressure Z-Spray ion source of a SYNAPT G2 operated at a source block temperature of 80 °C.

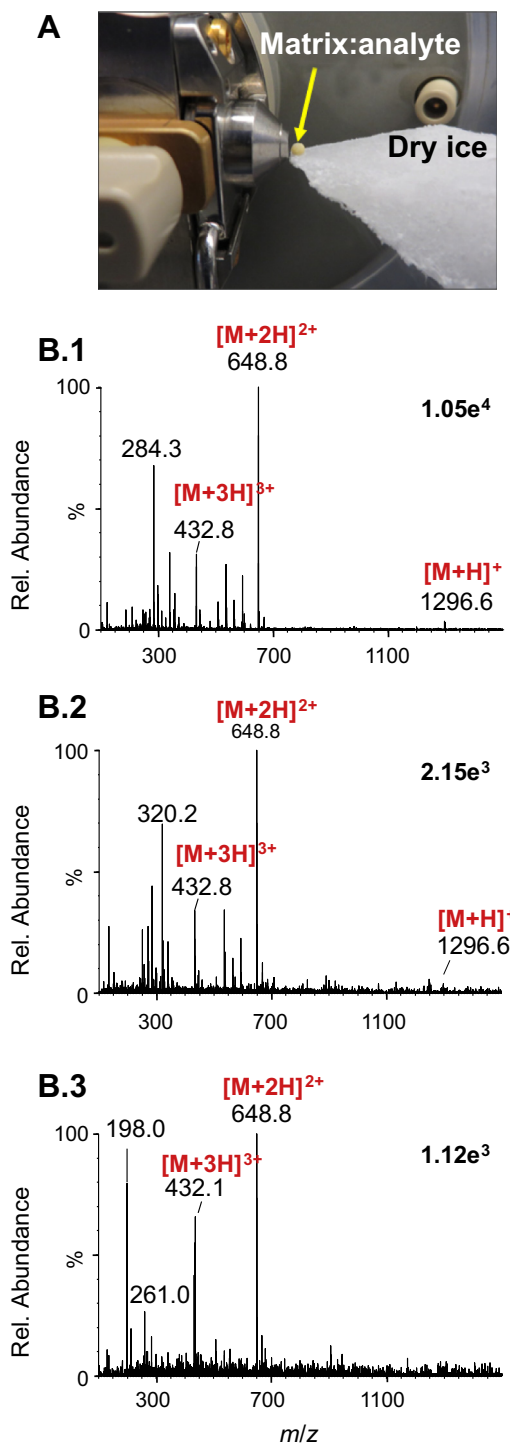


Fig. 2. (A) Photograph and (B) mass spectra of angiotensin I using representative liquid matrices (1) 1,3-dimethyl-2-nitrobenzene (L5), (2) 3-methoxybenzonitrile (L10), and (3) methyl-4-methyl-4-nitropentanoate (L13). Matrix–analyte samples were frozen onto dry ice (−80 °C) before introduction to the atmospheric pressure Z-Spray ion source of the SYNAPT G2 operated at a source block temperature at 30 °C. Additional mass spectral data of liquid matrix compounds can be found in Fig. S1.

crystalline matrix–analyte mixture (Fig. 1A), in solution (Fig. 1B), or frozen on dry ice either as crystals (Fig. 1C) or in solution (Fig. 1D). Under all conditions, methyl-5-nitro-2-furoate produces up to $[M+8H]^{8+}$ charge state ions in good abundance at the lowest available source block temperature of 30 °C. Clearly, numerous parameters need to be tested before a compound can be ruled out as a potential MAIV matrix; a limited set are described below.

Potential matrix compounds can be in the liquid or solid phases. Frequently, volatile compounds at room temperature are liquids. However, none of the compounds tested acted as MAIV matrices when directly introduced as liquids. Previous results using 450 °C inlet tube temperature also produced no analyte ions using liquid matrices [13]. Frozen solid matrices [7] and freezing water and methanol solutions [70] have been shown to produce analyte ions. We therefore used the atmospheric pressure pipet method, but froze the liquid matrices (Scheme 1, Table S1) using dry ice as described above (Fig. 1) for methyl-5-nitro-2-furoate (S24). In this approach, the liquid matrix containing the analyte was drawn into a pipet tip and brought in contact with dry ice to freeze the mixture. Care was taken that the aqueous solution of the analyte was removed prior to freezing the matrix–analyte to obtain the solid state crystal. The dry ice serves as sample support to introduce the solidified matrix–analyte sample to the Z-Spray inlet aperture operated at 30 °C (Fig. 2A). Care is also taken to avoid melting during sample introduction by having a small gap between the solid matrix–analyte and the inlet aperture. Cracking was observed as the matrix froze splintering off matrix–analyte which entered the inlet. Under freezing conditions, 15 additional compounds that are liquids at room temperature acted as MAIV matrices (L1–L15, Scheme 1) producing ions from incorporated analyte. Representative liquid matrices 1,3-dimethyl-2-nitrobenzene (L5), 3-methoxybenzonitrile (L10), and methyl-4-methyl-4-nitropentanoate (L13) are found to ionize the peptide angiotensin I with good ion intensity and produce up to $[M+3H]^{3+}$ charge state ions (Fig. 2B). Mass spectra for the additional 12 liquid matrices (L1–4, L6–9, L11, L12, L14, L15, Scheme 1) can be found in Fig. S1.

The freezing results are a good example of the potential for a compound to spontaneously ionize analyte under certain conditions. Another example is provided for a compound known

to exist in more than one crystalline form. Franco et al. reported that crystallization experiments of 2,5-diphenyl-1,3,4-oxadiazole (S22) from ethanol and petroleum ether as solvents revealed that the plate-like crystal structure rather than the needle-like structures exhibits triboluminescence and sublimates when exposed to temperatures above 90 °C [66]. For the studies reported here, the matrix was recrystallized twice in the respective solvents to obtain each form in high purity (Fig. 3A.1 and B.1). The solvent-free sample preparation method [69] was then utilized to mix the dried analyte angiotensin I with each matrix and preserve the crystal morphologies. The solvent-free sample preparation approach was previously shown to produce analyte ions using the 3-NBN matrix [7]. Only the sample prepared with plate-like crystal structures (Fig. 3A.1), reported to triboluminesce [66], produced abundant multiply charged ions of angiotensin I (Fig. 3A.2) at a source block temperature of 80 °C. No analyte ions were observed (Fig. 3B.2) from the needle-like crystals, reported not to triboluminesce [66]. Matrix crystals taken directly out of the bottle (Fig. 3C.1) produces only a protonated matrix peak at a mass-to-charge (m/z) of 223, but no analyte ions (Fig. 3C.2) when prepared solvent-free. These results are in agreement with the hypothesis that triboluminescence is an important component of charge separation in MAIV [7].

Because crystal morphology can influence the ionization process, we examined several matrices which provide either poor analyte ion intensity or metal cation adduction to analyte ions in order to determine if purification by crystallization might alter their effectiveness as a MAIV matrices. When the matrices, 5-bromo-3-nitropyridine-2-carbonitrile (S17), 3-methyl-4-nitroaniline (S6), 1,4-dihydroxy-2,6-dimethoxybenzene (S15), and 5-nitroindole (S18), which exhibit metal cation adduction from Cu^+ or K^+ , were recrystallized and separately mixed with angiotensin I, mass spectra (Fig. S2A–C) with higher signal-to-noise as well as less metal attachment, and often with notably higher abundance of the $[M+2H]^{2+}$ and $[M+3H]^{3+}$ ions, was obtained. However, some matrix compounds showed reduced ion intensity upon recrystallization. For example, phthalic anhydride (S25), 2'-hydroxy-5'-nitroacetophenone (S14), and 4-hydroxy-3-nitrocoumarin (S23) produced less abundant analyte ions, or none at all (Fig. S2D–F). It will be of interest to determine if purification affects triboluminescence in these compounds [71].

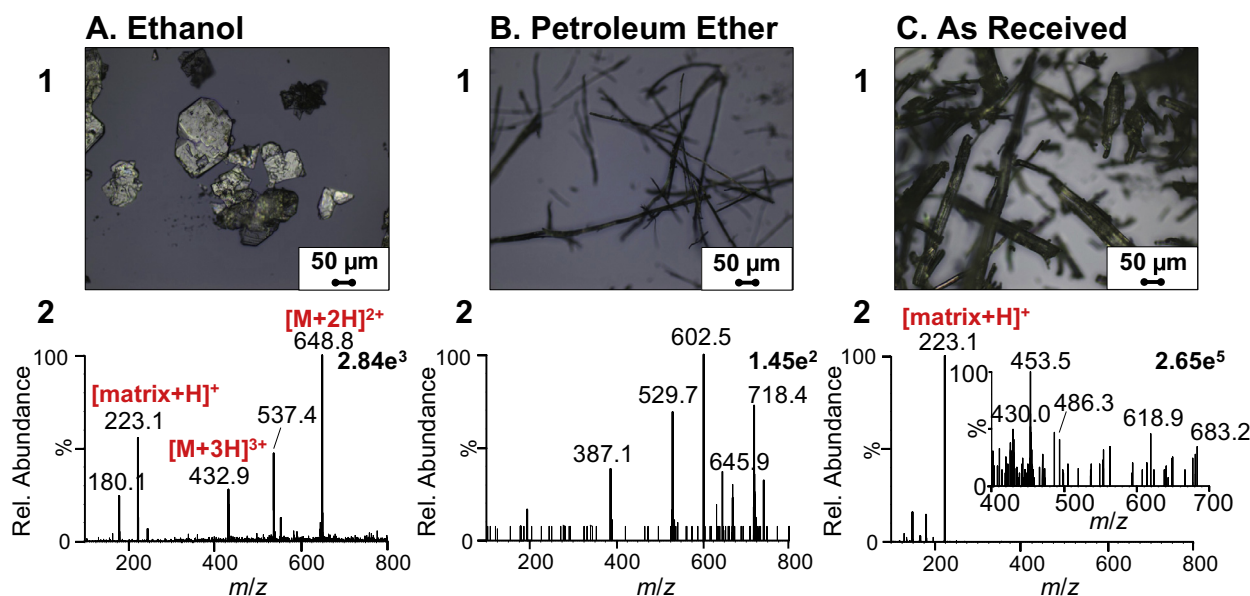


Fig. 3. Matrix 2,5-diphenyl-1,3,4-oxadiazole (S22) used after recrystallization from (A) ethanol or (B) petroleum ether as well as (C) without recrystallization as received: (1) microscopy image of crystals, and (2) corresponding MAIV mass spectra. The matrix–analyte (angiotensin I) sample preparation was by the solvent-free approach. The atmospheric pressure Z-Spray ion source of the SYNAPT G2 was operated at a source temperature of 80 °C.

The time over which the matrix sublimates relates to how long ions are observed and can be influenced by temperature (Fig. S3). With the atmospheric pressure Z-Spray source, heat from the source block can be applied to speed the rate of sublimation and momentarily increase the analyte ion abundance. At a source block temperature of 30 °C, matrix compounds 5-methyl-2-nitrobenzonitrile (S7), phthalonitrile (S9), methyl-5-nitro-2-furoate (S24), and 4-methyl-phthalonitrile (S10) sublime in under 4 min, while methyl-2-methyl-3-nitrobenzoate (S8), 4-aminophthalonitrile (S11), bronopol (S12), 1,4-dihydroxy-2,6-dimethoxybenzene (S15), and 2-naphthol (S16) sublime over a time range of 8 to 25 min (Fig. S3). Surprisingly, 6-nitro-*o*-anisonitrile (S5) required over 2 h to sublime at a source block temperature of 30 °C, while still producing good ion abundance. The general commonality between all MAIV matrices is an increased rate of sublimation at source block temperatures of about 80–150 °C (Fig. S3), where the ion signal lasts only for a few seconds. On the intermediate pressure vacuum source of this mass spectrometer, sublimation requires minutes, and in some cases, hours, likely influenced by less airflow, increased vacuum and/or operation from room temperature (Fig. S4).

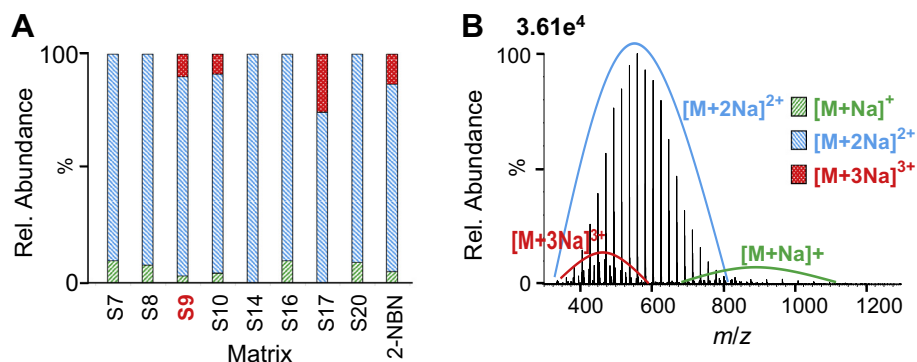
Interestingly, 2,5-DHAP, a common MALDI matrix [40] (Scheme 1IA), slowly sublimates under intermediate pressure conditions of the SYNAPT G2 MALDI source. By acquiring data over ~12 h, multiply charged ions were observed from the incorporated peptides leucine enkephalin, angiotensin I and II, and even up to the $[M+5H]^{5+}$ charge state of bovine insulin without use of a laser (Fig. S5A). Therefore, 2,5-DHAP, acts as a MALDI [27], LSII [72,73], LSIV [19], and MAIV matrix (from both atmospheric [7] and intermediate pressure). The matrix 2,5-DHB, a

MALDI [25] and LSII [10] matrix (Scheme 1IA), does not visually sublime from the surface and does not produce analyte ions from the peptide and protein mixture in the intermediate pressure vacuum source after acquiring data overnight (~12 h, Fig. S5B). The time scale of sublimation of 2,5-DHB may be too slow, in the absence of heat, to produce ions by this approach. CHCA (Scheme 1IA), another common MALDI matrix [13,18], is not observed to sublime over a period of ~12 h and also does not produce analyte ions (Fig. S5C). These results further suggest a mechanistic commonality between MALDI and the inlet and vacuum ionization methods as previously proposed [18].

While MAIV appears to depend on matrices that sublime in a reasonable time frame when placed in vacuum at a certain temperature, MALDI has been optimized to avoid matrices that sublime too quickly [24,27,29,30,74]. Because the new matrix-assisted ionization processes are pressure and temperature driven, we might expect that for each matrix, the pressure/temperature conditions for optimum ionization are different. As found in the MAIL study [13], many more matrices produce analyte ions at a high inlet temperature than at a low inlet temperature. Similarly, a heated LSIV source on a converted Z-Spray source extended the mass range and applicability to new compounds classes [75] relative to the room temperature operated intermediate pressure LSIV [19]. By extension, it is anticipated that heating either the sample holder or providing energy for desolvation will expand MAIV under vacuum conditions to a wider array of matrices.

While the performance of each matrix varies greatly, it is astonishing that 40 new matrix compounds have now been discovered that lift volatile and nonvolatile analyte molecules into the gas phase as ions when exposed to sub-atmospheric pressure

I. Intermediate Pressure



II. Atmospheric Pressure

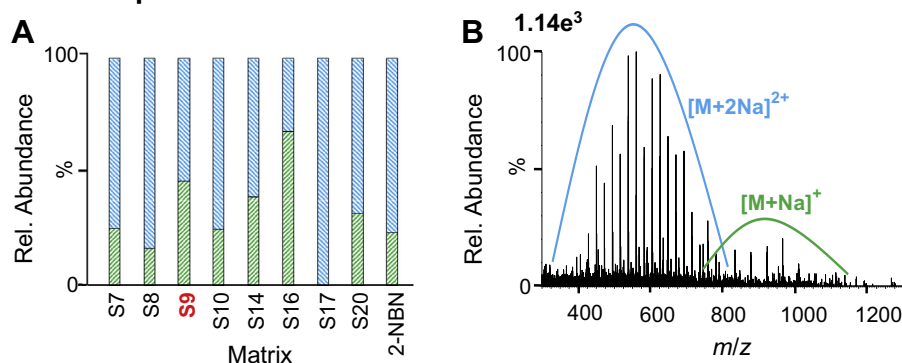


Fig. 4. Distributions of charge states $[M+Na]^+$, $[M+2Na]^{2+}$, and $[M+3Na]^{3+}$ for PEG 1000 using novel matrix compounds at (I) intermediate pressure (~0.16 Torr) and (II) atmospheric pressure. (A) Graphs of matrix compounds plotted against the relative ion abundance of respective charge states. (B) Representative mass spectra of PEG 1000 using the matrix phthalonitrile (S9) indicated in bold red. All data was acquired at ambient room temperature using a SYNAPT G2. Relative ion abundances can be found in Supplemental Fig. S6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

at moderate or even cold conditions. As discussed in more detail below, some of the new matrix compounds produce results with similar spectral quality and high ion intensity as 3-NBN, the first MAIV matrix discovered. Some newly discovered matrix compounds expand to new compound classes. Of these new matrices, the 25 that are solids at room temperature (S1–S25, Scheme 1II) are the basis of more detailed studies discussed below. In the case of triboluminescence, it is hypothesized that the crystal structure must be asymmetric [21] and these structures comply with this general characteristic.

3.2. Fundamental studies of MAIV matrices to enhance applications

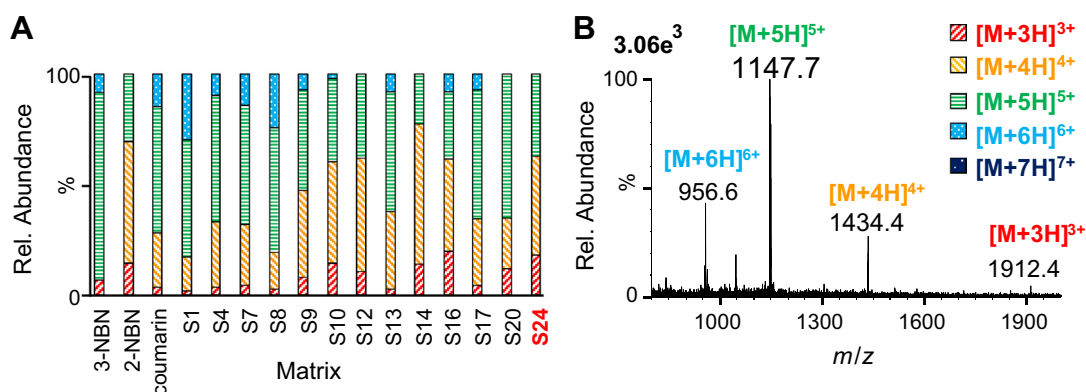
We proposed [7,18] that gas-phase analyte ion formation in the inlet and vacuum ionization methods occur by a similar process as in ESI [6]. However, this ignores the fact that ESI produces ions from solutions and MAIV seems to prefer ions be produced from the solid state. Likewise, it is tempting to consider MAIV as a continuation of MAIL, but the heat applied in MAIL provides additional opportunities for producing charge separation not available at room or low temperature used in MAIV [6–9,23]. Improvements in our fundamental understanding of this new ionization process should enable better matrix and instrument design to achieve higher sensitivity and broader or more selective ionization, and likely provide insights into mechanistic aspects of traditional ionization methods, offering additional impact. Finding additional MAIV matrices described above is a step in increasing our understanding of ionization processes. Below we describe pressure, temperature, and pH studies to provide additional

insights. Matrices studied are the 25 (S1–S25, Scheme 1II) that are solids at room temperature, operational with MAIV on atmospheric pressure Z-Spray sources. The results are compared with those acquired using known MAIV (Scheme 1IB) and MALDI matrices (Scheme 1IA).

3.3. Pressure studies

Previous studies using either matrix-assisted ionization or laserspray ionization showed straightforward applicability for peptides and proteins on the Z-Spray and intermediate pressure sources. Synthetic polymers and lipids ionized poorly, if at all, on the temperature limited Z-Spray source while working well on the hot inlet tube source of Thermo Scientific mass spectrometers [11–13,19,76,78]. Matrices that extended laserspray ionization to high vacuum MALDI-time-of-flight mass spectrometers are thus far only applicable for biological polymers [13,18,76]. A solution containing PEG 1000 (Figs. 4 and S6) was examined with the 25 new MAIV matrices (S1–S25) as well as the known MAIV matrices on both the atmospheric pressure Z-Spray source and the intermediate pressure (~0.16 Torr) vacuum source of the same SYNAPT G2 mass spectrometer. With the addition of 10 times molar excess of NaTFA salt, 8 of the 25 matrix compounds produced multiply charged ions of PEG at both intermediate (Fig. 4I) and atmospheric pressure (Fig. 4II) at 30 °C and ambient temperature, respectively. In general, an increase in ion abundance is observed at intermediate pressure relative to atmospheric pressure. For example, using phthalonitrile (S9) as the matrix, the ion abundance for the $[M+2H]^{2+}$ series increased by an order of magnitude relative to ionization from

I. Intermediate Pressure



II. Atmospheric Pressure

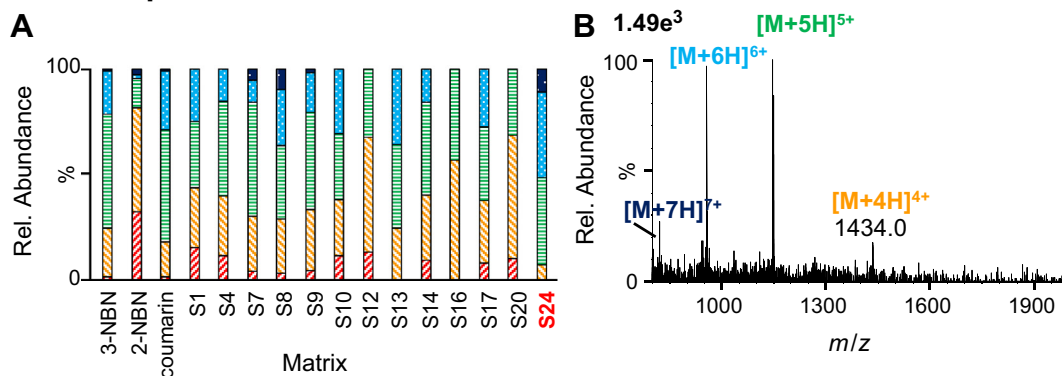


Fig. 5. Distributions of charge states $[M+3H]^{3+}$ through $[M+7H]^{7+}$ for bovine insulin (MW 5735) using novel matrix compounds at (I) intermediate pressure (~0.16 Torr) and (II) atmospheric pressure. (A) Graphs of matrix compounds plotted against the relative ion abundance of respective charge states. (B) Representative mass spectra of bovine insulin using matrix methyl 5-nitro-2-furoate (S24) indicated in bold red. All data was acquired at ambient room temperature using a SYNAPT G2. Relative ion abundances can be found in Supplemental Fig. S7 (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

atmospheric pressure (Fig. 4B). Additionally, three of the newly discovered matrix compounds, 5-bromo-3-nitropyridine-2-carbonitrile (S17), phthalonitrile (S9), 4-methyl-phthalonitrile (S10), as well as the known MAIV matrix, 2-NBN, produced up to triply charged $[M+3Na]^{3+}$ ions of the PEG oligomers (Fig. 4IA). The higher charge states observed at intermediate pressure relative to sample introduction from atmospheric pressure is unexpected and is may be related to more appropriate pressure conditions for these types of salty samples.

For bovine insulin, using the same source conditions, 13 of the 25 new MAIV matrices, and all previously known MAIV matrices, produced multiply charged ions at both intermediate (Fig. 5I) and atmospheric pressure (Fig. 5II). All 8 matrices that produced ions for PEG 1000 also produced ions for insulin. The difference between vacuum and atmospheric pressure conditions are less pronounced for bovine insulin than for PEG. For the bovine insulin study, perhaps the more apparent differences are some of the high

charge states observed. For example, methyl 5-nitro-2-furoate (S24) favors high charge states of $[M+7H]^{7+}$, $[M+6H]^{6+}$, and $[M+5H]^{5+}$ at atmospheric pressure relative to intermediate pressure where $[M+5H]^{5+}$ is the highest charge state observed. More importantly, this matrix tends to produce a higher abundance of the $[M+7H]^{7+}$ charge state when compared to matrix compounds methyl-2-methyl-3-nitrobenzoate (S8), phthalonitrile (S9), 4-methyl-phthalonitrile (S10), and even 3-NBN (Fig. 5IA).

Considering the previous challenges of the Z-Spray source for the analyses of synthetic polymer samples, and the trends observed here for both synthetic and biological polymers, these results suggests a fundamental role of pressure in the ionization process. Future studies will need to expand on the generality and optimize for maximum usefulness. We fully expect that other mass spectrometers perform differently because of differences in source designs and vacuum/temperature applicability.

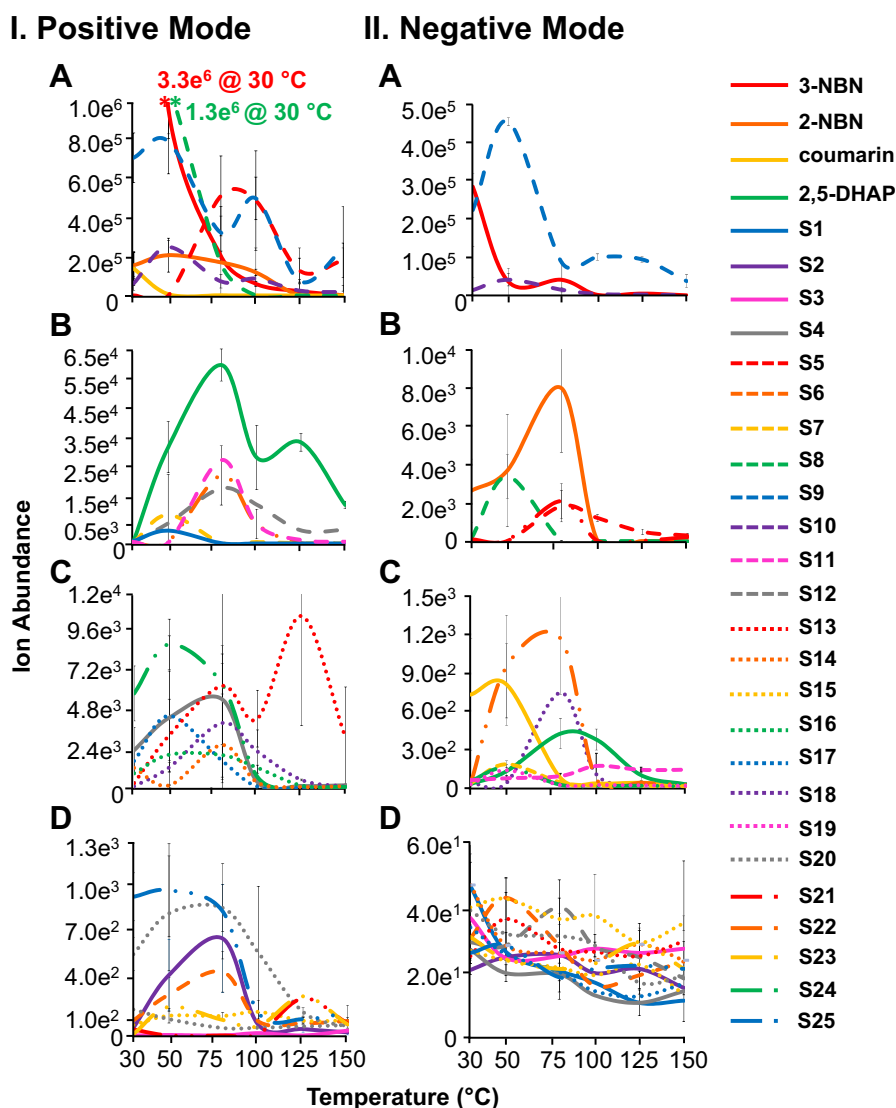


Fig. 6. Temperature versus ion abundance of (I) the $[M+5H]^{5+}$ charge state of bovine insulin in positive ion mode and (II) the $[M-2H]^{2-}$ charge state of GD1_b ganglioside lipid in negative ion mode. Matrix compounds are displayed in order of decreasing ion intensity: (IA) excellent: S5, S8, S9, S10, 3-NBN, 2-NBN, coumarin (IB) good: S1, S7, S11, S12, S22, 2,5-DHAP, (IC) mediocre: S4, S13, S14, S16, S17, S18, S24 (ID) poor: S2, S3, S6, S15, S19, S20, S21, S23, S25 (IIA) excellent: S9, S10, 3-NBN (IIB) good: S5, S8, S21, 2-NBN (IIC) mediocre: S7, S11, S18, S19, S22, S24, 2,5-DHAP, coumarin (IID) poor: S1, S2, S3, S4, S6, S12, S13, S14, S15, S16, S17, S18, S20, S23, S25. Error bars are shown for triplicate measurements of the respective ion abundances. Corresponding mass spectral data can be found in Supplemental Fig. S8 and Supplemental Fig. S9. Data was acquired using the Z-Spray ion source of a SYNAPT G2.

3.4. Temperature studies

For the 25 (S1–S25) newly discovered solid matrices operational on the Z-Spray source and 4 previously reported MAIV matrices (Scheme 1IB) and 2,5-DHAP (Scheme 1IA), we examined ion abundance for the $[M + 5H]^{5+}$ charge state of bovine insulin in the positive mode (Fig. 6I) with source block temperatures of the Z-Spray source at 30, 50, 80, 100, 125, and 150 °C. The $[M - 2H]^{2-}$ charge state of ganglioside GD1_b lipid in negative ion mode (Fig. 6II) was also examined at the same respective temperatures. The temperature studies for positive and negative modes are displayed according to excellent (Fig. 6A) to poor (Fig. 6D) performance of matrix compounds based on relative analyte ion intensity. The error bars for triplicate measurements are rather large showing that the methods of matrix–analyte introduction by pipet tip and/or glass plate (Table S1) currently have significant variation. Mass spectral data for all 25 solid matrices can be found in the supplemental information in Figs. S8 and S9.

As a baseline for comparison in the positive detection mode using the converted Z-Spray source, 3-NBN matrix gave the highest analyte ion abundance at the lowest temperature studied (30 °C), in accord with the excellent performance on intermediate pressure vacuum sources [6–8]. Nearly all the matrix compounds in this study display a significant decrease in ion abundance at elevated temperature, contrary to previous investigations with *inlet* ionization [10–18]. Interestingly, the optimal source block temperature is well below 150 °C for all of the selected matrix compounds in both positive and negative modes. Known MALDI matrices 2,5-DHB, dithranol, and CHCA (Scheme 1IA) produced no analyte ions at any of the temperatures studied. The novel MAIV matrices produce ions over a broad temperature range. For example, the matrices 6-nitro-*o*-anisonitrile (S5), phthalonitrile (S9), and 4-methyl-phthalonitrile (S10) produce high analyte abundance at all temperatures from 50 to 125 °C. Methyl-2-methyl-3-nitrobenzoate (S8) rivals the ion intensity of 3-NBN and provides exceptional ion abundance at temperatures of 30 and 50 °C in positive ion mode (Fig. 6IA).

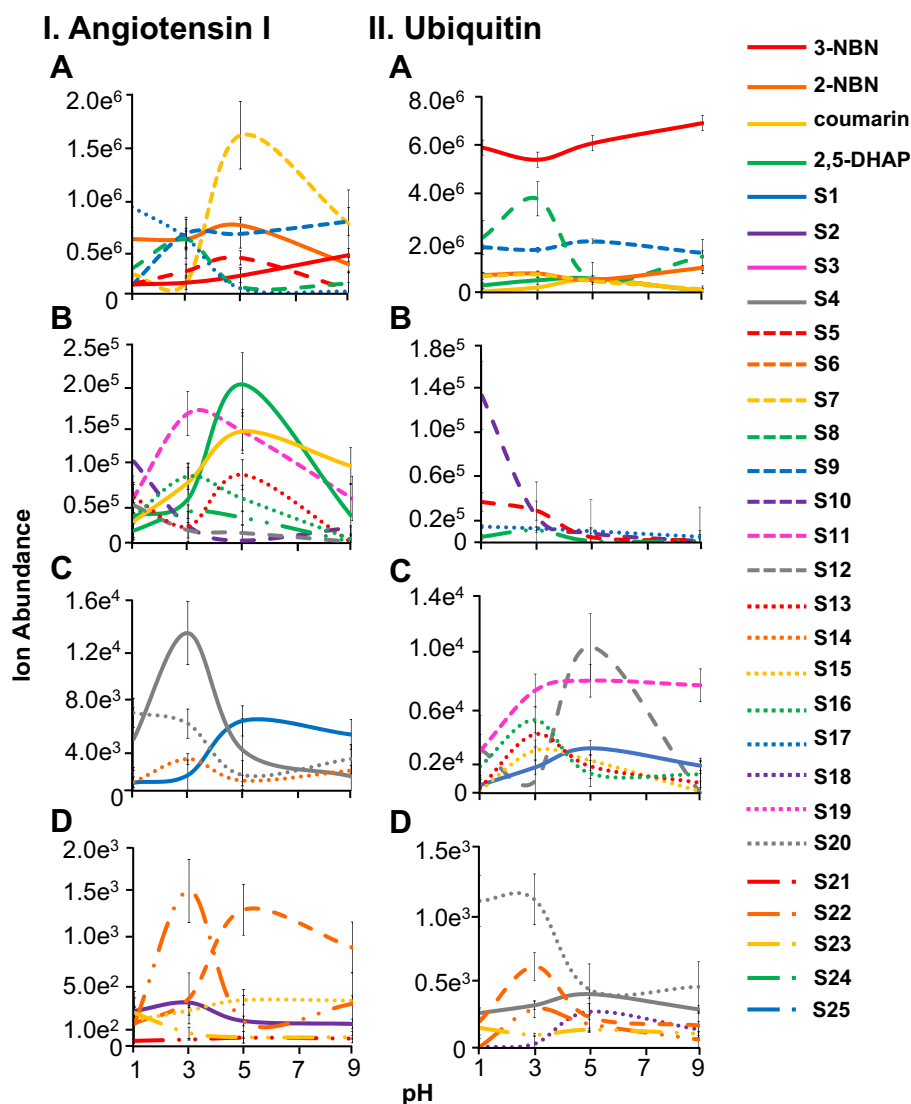


Fig. 7. pH versus ion abundance of (I) the $[M + 2H]^{2+}$ ion of angiotensin I and (II) the $[M + 8H]^{8+}$ charge state of ubiquitin in the positive mode. Matrix compounds are displayed in order of decreasing ion intensity: (IA) excellent: S5, S7, S8, S9, S17, 2-NBN, 3-NBN, (IB) good: S10, S11, S12, S13, S16, S24, 2,5-DHAP, coumarin, (IC) mediocre: S1, S4, S14, S20 (ID) poor: S2, S6, S15, S21, S22, S23. (IIA) excellent: S7, S8, S9, 2-NBN, 3-NBN, 2,5-DHAP, coumarin, (IIB) good: S5, S10, S17, S24, (IIC) mediocre: S1, S11, S12, S13, S15, S16, (IID) poor: S4, S6, S18, S20, S22, S23. Error bars are shown for triplicate measurements of the respective ion abundances. Corresponding mass spectral data can be found in Supplemental Fig. S10 and Supplemental Fig. S11. Data not shown for matrix compounds S2, S3, S14, S18, S19, S21, S25. The optimal source block temperatures for each respective matrix were used (see Fig. 6). Data was acquired using the Z-Spray ion source of a SYNAPT G2.

Importantly, some of the top performing matrix compounds in positive mode, phthalonitrile (S9), 4-methyl-phthalonitrile (S10), and 3-NBN (Fig. 6IA), provide high analyte ion abundance in the negative mode (Fig. 6IIA). All of the matrices that ionize insulin in the positive mode, however, were not found to be useful in ionizing gangliosides in the negative ion mode (Fig. 6II). Similar to MALDI [24–48], matrices are likely to be selective for different compound types and optimize under different conditions. A combination of temperature and pressure (Figs. 4–6) is likely to be effective in manipulating the duration and intensity of the analyte ions. The ability to form negatively charged analyte ions with the new MAIV matrices is in agreement with the hypothesis of sublimation driven triboluminescence [6,7]. Charge separation by the proposed mechanism active in triboluminescence creates as many positive as negative charges [57–61].

3.5. pH studies

The matrix 3-NBN, as well as a number of other MAIV matrices (Scheme 1), have no exchangeable protons, suggesting that analyte is pre-charged in solution and remains pre-charged in the solid matrix in order to produce the multiply charged gas-phase ions. It might be expected that matrices such as the isomers 2-NBN, 3-NBN, and 4-NBN would behave similarly as a function of pH, and matrices such as 2,5-DHAP with exchangeable hydrogens may have a different analyte response as the pH is changed. Clearly, pH is not the most critical factor, as 4-NBN does not assist in ionization under any of the conditions used so far, nor did any other compounds without a dipole moment [7]. However, a pH study provides at least information on a general trend for the best pH for any given analyte and matrix to help make intelligent decisions when acquiring data.

The ion abundances during the initial pH screening study varied depending on the matrix compound used at the optimal temperature shown in Figs. 7 and S10–11. To determine the influence of pH, these studies used a solution of angiotensin I (MW 1295) and ubiquitin (MW 8565) which are known to ionize well with 3-NBN. Both analytes were therefore dissolved in 50:50 methanol:water and acidified with 0.1% hydrochloric acid (~pH 1), 0.1% formic acid (~pH 3), at the natural pH of the solvent (~pH 5), and 0.1% ammonium hydroxide (~pH 9), respectively. The pH study was performed for all 25 novel matrices (S1–S25) that are solids at room temperature (Scheme 1II) and compared against four known MAIV matrices (Scheme 1IB) and 2,5-DHAP (Scheme 1IA). Fig. 7 displays the pH versus the average ion abundance of triplicate measurements for the doubly charged ion of angiotensin I and the $[M+8H]^{8+}$ charge state for ubiquitin. The matrices are ranked and displayed in order of exceptional (Fig. 7A) to poor performance (Fig. 7D). Source block temperature was set to the optimal temperature for each matrix as determined by the temperature studies (Fig. 6).

For the peptide (Fig. 7I), most of the novel matrices produced the highest ion abundance at the natural pH of 5 of the 50:50 methanol:water solution (Figs. 7IA and IB). Matrices phthalonitrile (S9) and 3-NBN show increased ion abundance for this peptide at pH 9 relative to natural and especially acidic conditions. On the other hand, matrices methyl 2-methyl-3-nitrobenzoate (S8), 4-methyl-phthalonitrile (S10), and 5-bromo-3-nitropyridine-2-carbonitrile (S17) show a modest trend towards better results for more acidic pH. Matrix compounds 2-bromo-3'-nitroacetophenone (S3), 5-nitroindole (S18), 9-vinyl carbazole (S19), and phthalic anhydride (S25) produced little or no analyte ions for angiotensin I.

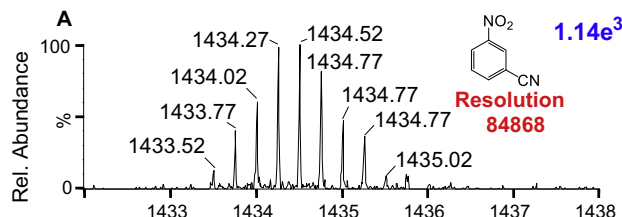
For ubiquitin (Fig. 7II), the preference for the natural pH is less apparent. The best performing matrices (Fig. 7IIA) worked well across the entire pH range of 1 to 9. Other matrices, especially those considered to be 'good' (Fig. 7IIB) show some preference for

highest ion abundance at acidic conditions of pH 1 and others at pH 3, e.g. methyl 2-methyl-3-nitrobenzoate (S8) and 4-methyl-phthalonitrile (S10), respectively. However, matrices 3-nitrobenzoyl chloride (S2), 2-bromo-3'-nitroacetophenone (S3), 2'-hydroxy-5'-nitroacetophenone (S14), 9-vinylcarbazole (S19), 3,6-dibromocarbazole (S21), and phthalic anhydride (S25) produced little or no ions from ubiquitin.

The results show that pH conditions influence the ion abundance of the peptide more than the protein as indicated by the large decrease in abundance at pH's other than that of the natural pH of the dissolved matrix compound (Fig. 7A). Some matrices behave similarly for peptides and proteins relative to the preferred pH value, e.g. methyl 2-methyl-3-nitrobenzoate (S8), 4-methyl-phthalonitrile (S10), and 2-naphthol (S16). Some of these matrices that performed poorly or not at all (Fig. 7C–D), only produced low abundant analyte ions at source block temperatures of 50 °C.

Clearly, pH plays a role in the ionization process and is related to the matrix compound and analyte used. However, we know that solution composition and crystallization of the matrix are also important factors in determining ion abundance, and thus, some of the pH study results might be explained by changes other than pH. It is expected that future studies will need to establish best

I. Intermediate Pressure Vacuum Source



II. Atmospheric Pressure Z-Spray Source

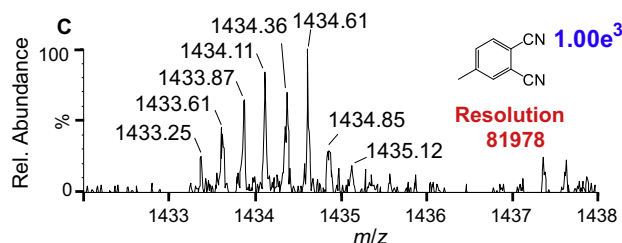
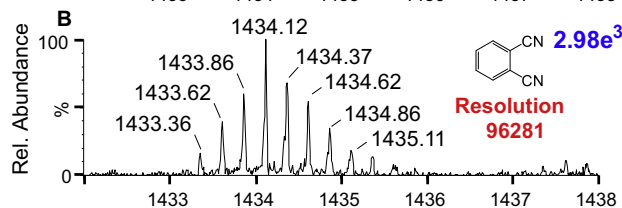
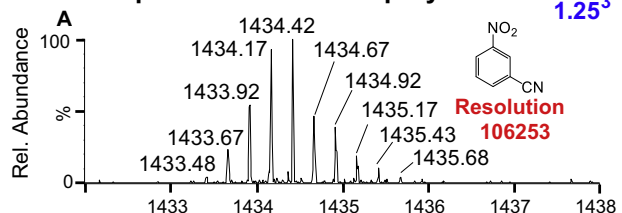


Fig. 8. Mass resolution values obtained in using the (I) intermediate pressure (~0.16 Torr) vacuum source at ambient temperature and (II) atmospheric pressure Z-Spray ion source at 50 °C of a SYNAPT G2 operated in high resolution mode. Zoom of the isotopic distribution of the $[M+4H]^{4+}$ ion using matrix compounds (A) 3-NBN, (B) phthalonitrile (S9), and (C) 4-methyl-phthalonitrile (S10).

performance dependent on pH and temperature variation for certain analytes.

3.6. Considerations relevant to solid matrices

Because the analyte charge states in MAIV are nearly identical to ESI, we can assume that there must be similarities in the ion formation mechanisms, similar to a previous proposal [18]. Without energy input other than that extracted from the matrix by the vacuum and under cold vacuum conditions, the MAIV matrices must remain in the solid state throughout the ionization process. Charged solid state particles would then be the MAIV equivalent of charged droplets hypothesized for ESI and *inlet* ionization, in which matrix–analyte particles melt upon ablation [17], similar to MALDI [79]. The hot inlet tube in MAII potentially also aids melting [18]. Thus, irrespective of the physical state of the charged matrix or solvent, charge states similar to ESI are observed. Further, drift times observed for laserspray and ESI [72] are similar as are the drift time distributions of ESI and MAIV ions using 3-NBN or the newly discovered matrices (Fig. S12).

Any proposed ionization mechanism needs to account for production of ions, especially multiply charged ions from large nonvolatile compounds, from liquid or solid particles with efficiency.

3.7. Applications using MAIV matrices

Though 3-NBN matrix has been a “gold-standard” for many MAIV analyses using a wide range of analytes [6,8,9]; it is not so effective with less basic compounds such as carbohydrates and synthetic polymers [7,77]. Other compounds such as ganglioside lipids have not been applicable using *inlet* or *vacuum* ionization because of insufficient heat on the Waters Z-Spray source and harsh conditions on the intermediate pressure source, respectively [78]. Through fundamental studies, a number of matrix compounds were found to consistently produce high analyte ion intensity with good spectral quality over a larger range of solvent, temperature, and source conditions for peptides, proteins, ganglioside lipids and PEG. On the atmospheric pressure Z-Spray source, 10 fmol of bovine insulin was detected using the matrix phthalonitrile (S9) (Fig. S13A). 5-Bromo-3-nitropyridine-2-carbonitrile (S17) and 4-methyl-phthalonitrile (S10) also demonstrated exceptional sensitivity, ionizing 50 fmol of insulin (Fig. S13B–C). On the intermediate pressure vacuum source, as low as 5 fmol of clozapine was detected in high ion abundance (Fig. S14) using the slower subliming matrix compounds methyl-2-methyl-3-nitrobenzoate (S8), phthalonitrile (S9), and 4-methyl-phthalonitrile (S10) prolonging the ionization event. Previous studies employing 3-NBN required the sample plate to be cooled to -80°C to prolong sublimation of the matrix/analyte under intermediate pressure and increase the sensitivity [7].

Out of 40 novel matrix compounds, 6-nitro-*o*-anisonitrile (S5), 5-methyl-2-nitro-benzonitrile (S7), methyl 2-methyl-3-nitrobenzoate (S8), phthalonitrile (S9), 4-methyl-phthalonitrile (S10), 2-bromo-2-nitro-1,3-propanediol (S12), 2-methyl-2-nitro-1,3-propanediol (S13), 2-naphthol (S16), 5-bromo-3-nitropyridine-2-carbonitrile (S17), and methyl 5-nitro-2-furoate (S24) have been observed to have sensitivities for certain analytes similar to 3-NBN. As examples, polymers (Fig. 4), carbohydrates (Fig. S15), and fragile lipids (Figs. 6 and S9) are now applicable demonstrating that new types of compound classes can be analyzed using the new MAIV matrices such as 5-bromo-3-nitropyridine-2-carbonitrile (S17) and phthalonitrile (S9), respectively. This suggests that the chemical structure, while having no specific elements required for ionization, is important in which compounds are preferentially ionized. While these studies are in the initial phases, we fully

expect, similar to MALDI [24–48], MAIV matrices will be specific to applications such as carbohydrate, polymer, peptide, protein, lipid, and small molecule analyses.

The matrix compounds presented herein, without the need for addition of heat or external voltages, provide good mass resolution for the $[\text{M}+4\text{H}]^{4+}$ ion of bovine insulin at m/z 1434. 3-NBN produces exceptional mass resolution from intermediate pressure (Fig. 8I.A) as well as from atmospheric pressure (Fig. 8II.A). Novel matrices that bear structural similarities, such as phthalonitrile (S9) (Fig. 8II.B) and 4-methyl-phthalonitrile (S10) (Fig. 8II.C) also produce high mass resolution for bovine insulin.

4. Conclusions

The finding that the 3-NBN used as a matrix lifts compounds from the solid state to gas-phase ions by simple exposure to low pressure conditions is now demonstrated, by the discovery of a number of new matrix compounds, to be a more general process than might have been anticipated. Liquid-phase matrices spontaneously produce analyte ions when exposed to vacuum of the mass spectrometer, but only when frozen possibly suggesting that splintering of charged particles from a solid surface is a necessary process for MAIV matrices. These MAIV matrices have a variety of structures suggesting that physical properties are more important than any particular chemical structure in the charge separation process. The results suggest that each matrix has a temperature as well as a preferred pressure range in which it most efficiently produces analyte ions. In this initial screening study, proteins and synthetic polymers show different preferences for pressure conditions. Contrary to MAII [13], the ionization temperature in MAIV is commonly well below 150°C . We envision preferential ionization of mixtures and designer matrices that are specific or inclusive, and foresee that slow ionization will improve the utility of MS/MS applications similar to LC-MS/MS [80], and fast ionization for high throughput applications will be made possible by simple temperature variation and matrix selection.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.ijms.2014.07.033>.

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